

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
O-LEVEL · COMBINED SCIENCE

COMBINED SCIENCE 5006/1

PAPER 1 November 2018

1 hour

PAPER 1 — MULTIPLE CHOICE — 40 QUESTIONS — 1 HOUR

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2018 5006/1.

Additional materials: Answer all 40 questions. For each question there are four possible answers A, B, C and D.

INSTRUCTIONS: Choose the one you consider correct and record it. Each question carries 1 mark.

1. A neutron has [1]

- A) charge –1
- B) charge +2
- C) no charge
- D) charge +1

2. Average speed is [1]

- A) distance × time
- B) distance / time
- C) time / distance
- D) mass / time

3. During electrolysis, cations move to the [1]

- A) cathode
- B) fuse
- C) anode
- D) voltmeter

4. An acidic solution has [1]

- A) pH 7
- B) pH 14 only
- C) pH less than 7
- D) pH greater than 7

5. Plasma transports [1]

- A) dissolved substances such as carbon dioxide, urea and hormones
- B) only starch grains
- C) only cellulose
- D) only oxygen in haemoglobin

6. Carbon dioxide in air is tested with [1]

- A) blue litmus only
- B) a lighted splint
- C) iodine
- D) limewater, which turns milky

7. Urea is made in the [1]

- A) skin from sweat only
- B) kidney from urine
- C) lungs from oxygen
- D) liver from excess amino acids

8. Ionic bonding is the [1]

- A) transfer of electrons from metal to non-metal, forming ions
- B) sharing of a pair of electrons
- C) attraction between nuclei only in metals
- D) hydrogen bonding in water only

9. Which material is magnetic? [1]

- A) wood
- B) plastic
- C) copper
- D) iron

10. A mass of 5 kg accelerates at 2 m/s^2 . The unbalanced force is [1]

- A) 50 N
- B) 0 N
- C) 10 N
- D) 7 N

11. Iron rusts when it is in contact with [1]

- A) nitrogen only
- B) argon
- C) oil only
- D) oxygen and water

12. Lipase, with bile, digests fats to [1]

- A) amino acids
- B) starch
- C) fatty acids and glycerol
- D) maltose

13. An enzyme is [1]

- A) a waste product of respiration
- B) a carbohydrate that stores energy
- C) a type of mineral ion
- D) a biological catalyst

14. A cellulose cell wall is found in [1]

- A) animal cells only
- B) viruses only
- C) both animal and plant cells
- D) plant cells only

15. Noble gases (Group 0/18) are [1]

- A) halogens
- B) alkali metals
- C) unreactive because they have a full outer shell
- D) very reactive metals

16. Ciliated cells in the trachea [1]

- A) pump blood
- B) sweep mucus and trapped particles
- C) make insulin
- D) absorb digested food

17. Density is [1]

- A) mass + volume
- B) mass $\times$ volume
- C) volume / mass
- D) mass / volume

18. Acid + carbonate produces [1]

- A) salt + hydrogen
- B) water only
- C) salt + oxygen
- D) salt + water + carbon dioxide

19. Length is measured with a [1]

- A) newton meter
- B) metre rule / tape
- C) stopwatch
- D) thermometer

20. The change from liquid to gas is [1]

- A) condensation
- B) freezing
- C) melting
- D) evaporation / boiling

21. The volume of a liquid is measured with a [1]

- A) metre rule
- B) beam balance only
- C) measuring cylinder
- D) stopwatch

22. Which statement is correct? [1]

- A) Conduction occurs only in gases
- B) Convection occurs in solids only
- C) Radiation does not need a medium
- D) Conduction is bulk movement of a fluid

23. The SI unit of force is the [1]

- A) newton
- B) joule
- C) kilogram
- D) watt

24. Aerobic respiration takes place mainly in the [1]

- A) cell wall
- B) nucleus
- C) chloroplast
- D) mitochondrion

25. The word equation for photosynthesis is **[1]**

- A) nitrogen + water → glucose + oxygen
- B) glucose + carbon dioxide → oxygen + water
- C) glucose + oxygen → carbon dioxide + water
- D) carbon dioxide + water → glucose + oxygen

26. A 4 Ω resistor is connected to a 12 V supply. The current is **[1]**

- A) 3 A
- B) 48 A
- C) 16 A
- D) 4 A

27. Salivary amylase begins the digestion of **[1]**

- A) protein to amino acids
- B) starch to maltose
- C) fibre to glucose
- D) fat to glycerol

28. Platelets are needed for **[1]**

- A) photosynthesis
- B) blood clotting
- C) making bile
- D) osmoregulation only

29. Sperm are produced in the **[1]**

- A) testes
- B) oviduct
- C) prostate only
- D) uterus

30. In double circulation, blood passes through the heart **[1]**

- A) once per circuit of the body
- B) only at night
- C) not at all
- D) twice per circuit of the body

31. The electronic configuration of chlorine (proton number 17) is **[1]**

- A) 2,6
- B) 2,8,7
- C) 2,8,2
- D) 2,8,1

32. A herbivore is a **[1]**

- A) pathogen
- B) producer
- C) primary consumer
- D) decomposer only

33. The pitch of a sound depends on its **[1]**

- A) frequency
- B) colour
- C) loudness only
- D) amplitude

34. Which is true of magnets? [1]

- A) North-south attract
- B) Like poles attract
- C) North-north attract
- D) South-south attract

35. Sound travels fastest in [1]

- A) a vacuum
- B) air
- C) water
- D) steel

36. For main-group elements, the group number indicates the [1]

- A) number of electron shells
- B) number of neutrons
- C) number of outer-shell electrons
- D) mass number

37. Neutral solutions have pH [1]

- A) 14
- B) 1
- C) 7
- D) 0

38. HIV is a [1]

- A) virus that attacks white blood cells / the immune system
- B) bacterium
- C) deficiency of vitamin C
- D) protozoan

39. A unit of density is [1]

- A) J
- B) W
- C) g/cm³
- D) N

40. Diffusion is fastest in a [1]

- A) liquid
- B) gas
- C) solid
- D) crystal lattice at 0 K

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).